

Cis-Regulatory Elements (CREs) in Diapause-Associated Genes

Comparative analysis, Human vs. Mouse | Python / Biopython / pandas / Matplotlib | project summary

The gist. Embryonic diapause - a reversible pause in embryo development - is controlled mainly by *how* genes are switched on and off, i.e. by cis-regulatory elements (CREs): short promoter motifs that transcription factors bind. This project asks whether the CRE wiring around diapause genes is shared between human and mouse.

Answer: the regulatory code is strongly conserved - per-motif frequencies match across species ($r = 0.95$ once one divergent element is set aside), the focal promoter is ~95% identical, and the same CRE pairs cluster together in both species. The one exception is **AP-1**, which has expanded specifically in the mouse lineage (~3x).

The question

Diapause is governed less by the genes themselves than by their regulatory switches. If two species as distant as human and mouse use the *same* CREs - at similar frequencies, positions, and combinations - that points to an ancient, conserved regulatory programme for dormancy. The analysis tests this directly by reading promoter DNA and quantifying its regulatory motifs.

Data and approach

Six diapause-linked genes are studied in both species: **MYC** and **MTOR** (growth / quiescence), **ESRRB** and **LIF** (pluripotency / implantation), **FOXO3** (dormancy and stress resistance), and **TP53** (dormancy checkpoint). By default the pipeline runs on a reproducible **synthetic dataset with a known ground truth** - this makes it run offline and, crucially, lets the motif scanner be validated before it is trusted. A single switch (USE_REAL_DATA) plus NCBI / Biopython and FASTA loaders run the identical pipeline on real promoter sequences. Four steps:

- **Parse and count** - scan each promoter for seven CRE motifs with overlap-aware regular expressions, normalised to occurrences per 1,000 bp so genes of different length compare fairly.
- **Compare** - frequency bar charts and gene x motif heatmaps across the two species.
- **Align** - a dot plot and a percent-identity for an orthologous pair (Biopython PairwiseAligner, with a built-in Needleman-Wunsch fallback).
- **Cluster** - find motif pairs that physically co-occur (within 30 bp) in both species: candidate conserved regulatory modules.

The CRE 'vocabulary' scanned

Motif	Consensus	Role / relevance to diapause
TATA box	TATAWAW	Core promoter - sets the transcription start
CAAT box	CCAAT	Core promoter - general activator element

GC box (Sp1)	GGGCGG	Core promoter - common in GC-rich promoters
E-box	CACGTG	Bound by MYC / bHLH; MYC is suppressed in diapause
FOXO	TTGTTTAC	FOXO drives dormancy and metabolic slow-down
CREB / CRE	TGACGTCA	cAMP-responsive; implantation / diapause exit
AP-1	TGACTCA	Stress response - here, a species-divergent element

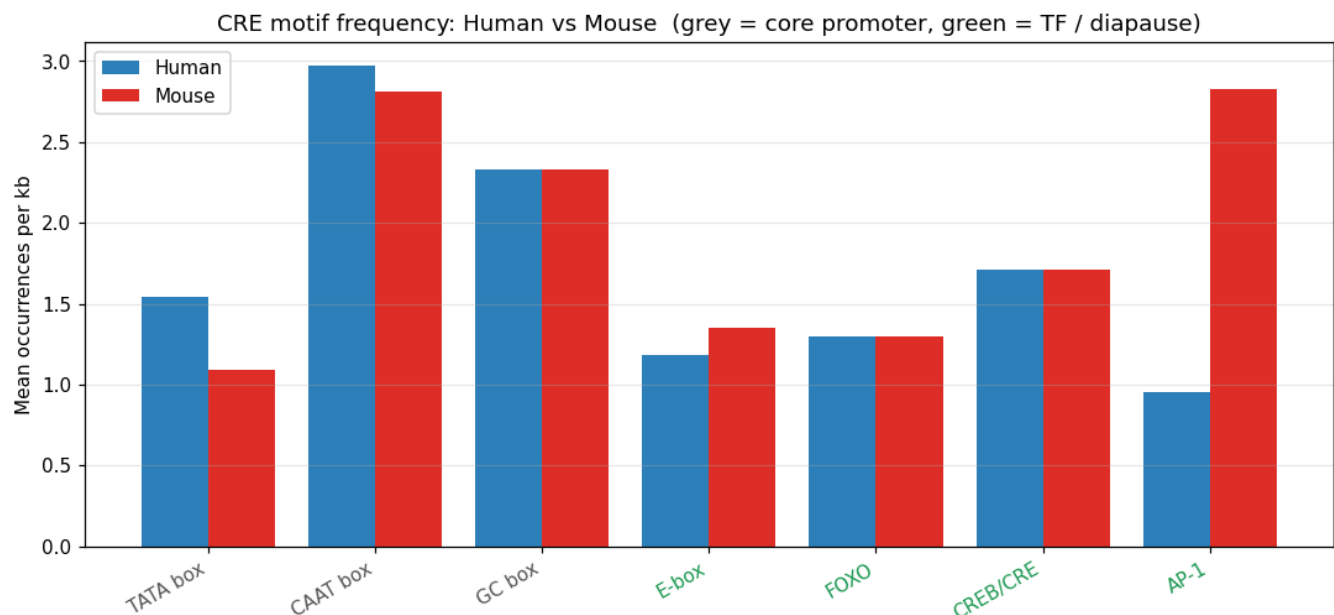
The first three are core-promoter elements expected in nearly every gene; the last four are transcription-factor CREs whose combination encodes gene-specific logic.

Results at a glance

Cross-species CRE conservation	$r = 0.84$ (all motifs); $r = 0.95$ excluding the divergent AP-1
Focal MYC promoter identity	95.0% (clear dot-plot diagonal + a 3-bp indel)
AP-1 abundance in mouse	~3x human (lineage-specific expansion)
Conserved co-occurring CRE pairs	13 found; strongest: FOXO+CAAT, E-box+CREB, GC+FOXO

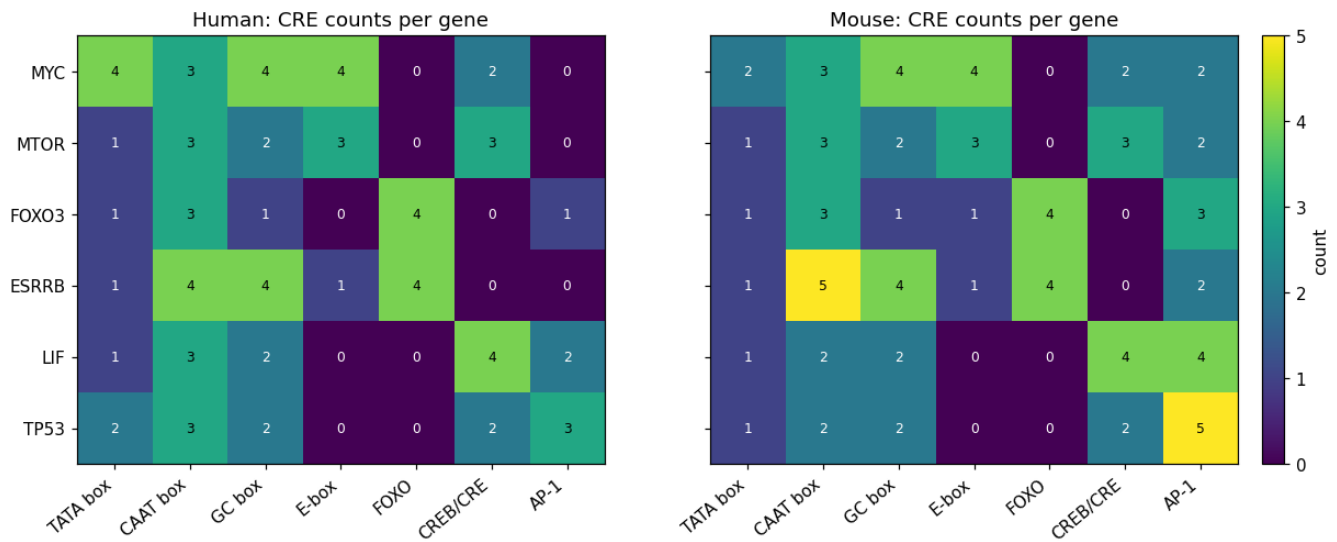
What the charts show

1. Motif frequency. Core-promoter and diapause-linked TF motifs occur at near-identical rates in both species; AP-1 stands out as much higher in mouse.



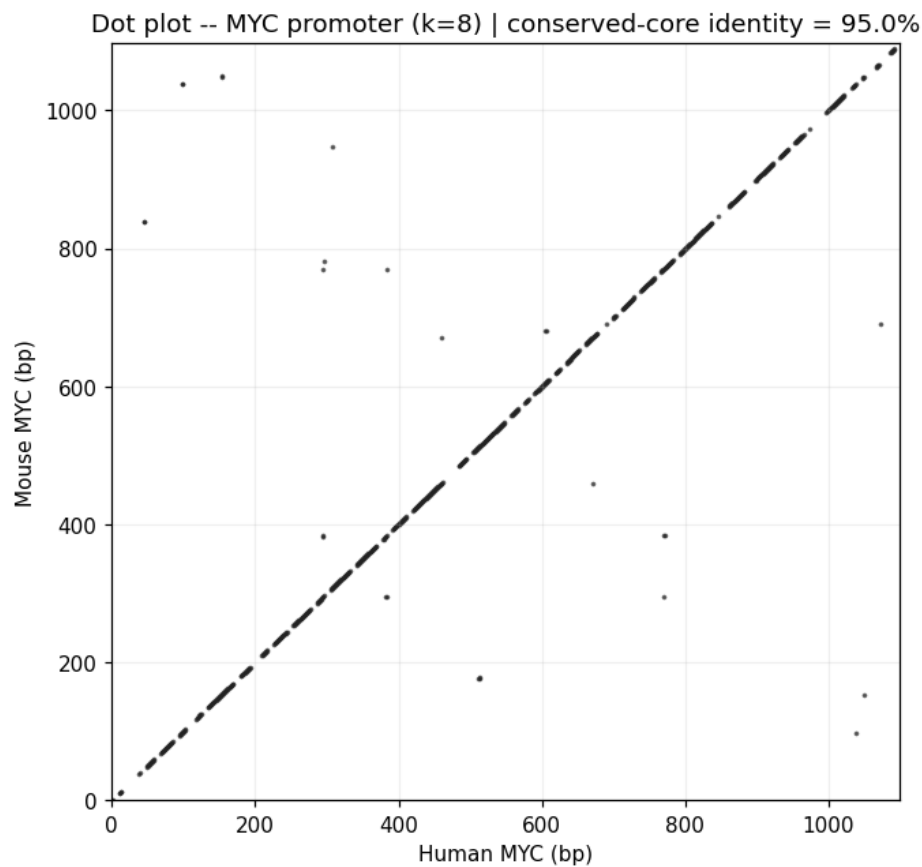
CRE motif frequency (mean per kb). Bars for the conserved motifs are essentially equal height between human and mouse; AP-1 (far right) is markedly higher in mouse.

2. Per-gene signatures. Each gene carries its own characteristic mix of CREs, and the human and mouse patterns line up closely.



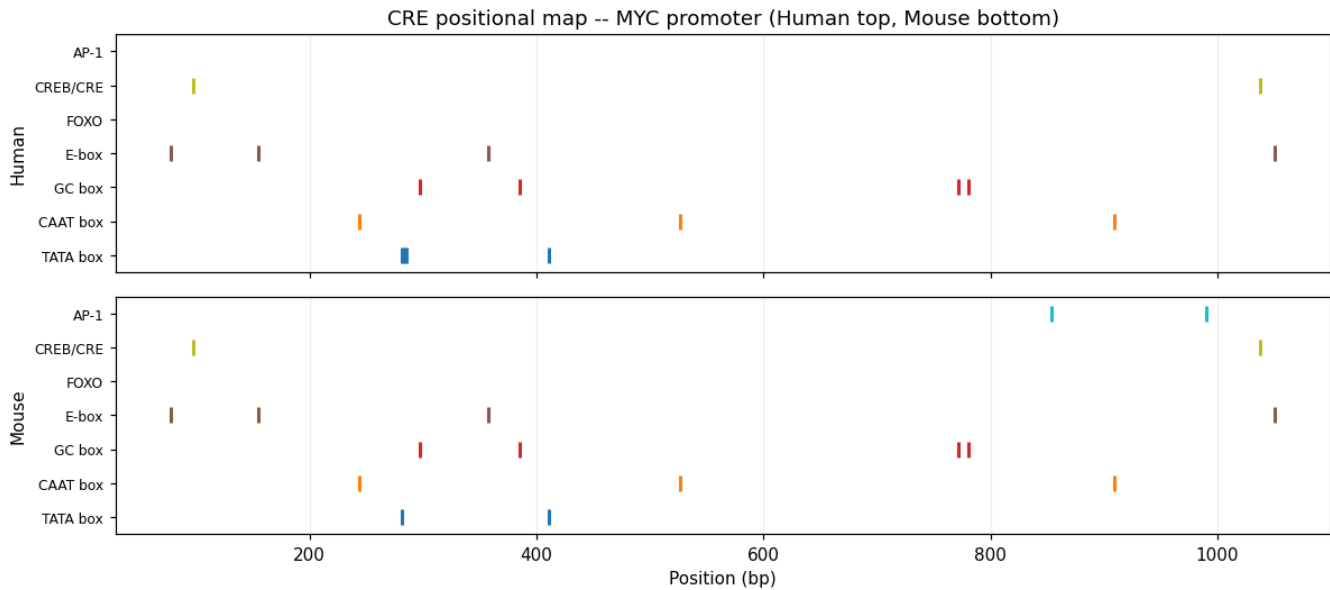
CRE counts per gene (left: human, right: mouse). The two heatmaps are visually almost identical - regulatory content is conserved gene by gene.

3. Promoter alignment. The orthologous MYC promoters are clearly homologous; a deliberate small deletion in mouse is visible as a break in the diagonal.



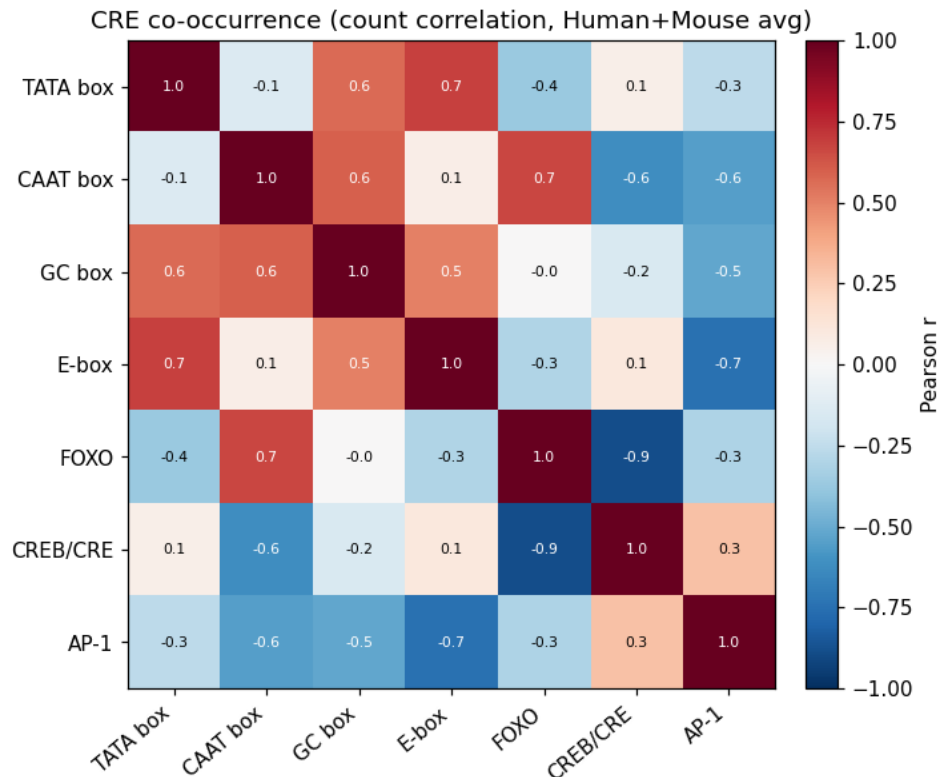
Dot plot of the MYC promoter (human vs mouse). A continuous diagonal indicates homology; the offset near the 3' end is the engineered 3-bp deletion. Windowed identity = 95%.

4. Positional conservation. Conserved CREs sit at matching positions in both species - and the mouse-specific AP-1 sites appear as extra ticks.



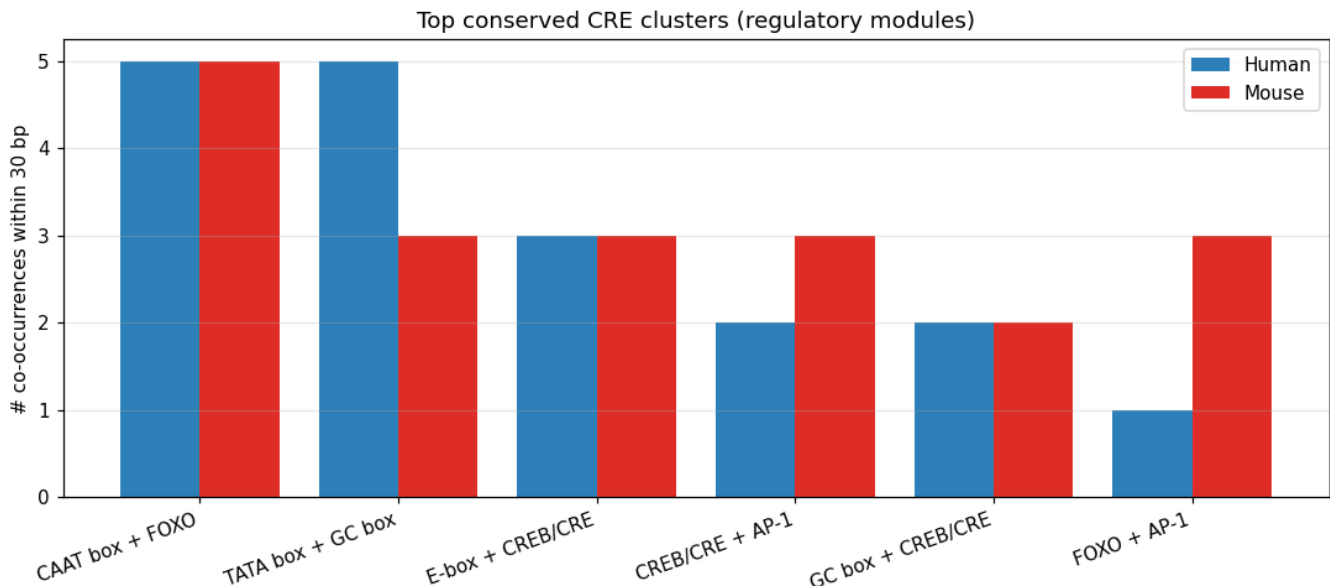
Positions of each CRE along the MYC promoter (human top, mouse bottom). Conserved motifs align position-for-position; the two extra AP-1 ticks in mouse are the lineage-specific expansion.

5. Co-occurrence structure. Some motif pairs reliably appear together across genes, the signature of combinatorial regulation.



CRE co-occurrence (count correlation across genes, averaged over species). Warm cells mark motif pairs that tend to be present together.

6. Conserved modules. Specific CRE pairs co-occur within short windows in both species. Non-AP-1 modules are consistent; AP-1-containing pairs are inflated in mouse.



Top conserved CRE clusters - pairs co-occurring within 30 bp in both species. FOXO+CAAT, E-box+CREB and GC+FOXO are consistent across species; AP-1-containing pairs are higher in mouse.

What it means

- **Conserved frequencies, one exception.** Setting AP-1 aside, human and mouse CRE frequencies match tightly ($r = 0.95$); core and diapause-linked motifs occur at similar rates.
- **Conserved sequence and positions.** The focal promoter is ~95% identical and CREs sit at matching positions in both species (a 3-bp mouse deletion is the only visible structural change).
- **Conserved regulatory modules.** The same CRE pairs (FOXO+CAAT, E-box+CREB, GC+FOXO) cluster together in both species - evidence that particular regulatory pairs are used as units.
- **Lineage-specific divergence.** AP-1 has expanded ~3x in mouse, a reminder that against a conserved background, some regulatory turnover still happens.

Limitations and next steps

The default dataset is synthetic, so the findings demonstrate the *method* rather than a new biological claim; motifs are matched as exact regex rather than probabilistic position-weight matrices; the gene panel is small; and the alignment is pedagogical rather than genome-scale. Natural extensions: plug in real Ensembl / NCBI promoters (one switch), score motifs with JASPAR PWMs (FIMO / MOODS), add statistical enrichment tests, and use multiple-sequence alignment or phylogenetic footprinting to call truly conserved CREs.

How to run and extend

Run the notebook

```
pip install jupyter biopython pandas matplotlib numpy  
jupyter notebook -> open the file -> Kernel -> Restart and Run All
```

Use real sequences

Set `USE_REAL_DATA = True` and supply RefSeq accessions (NCBI / Biopython Entrez) or promoter FASTA exported from Ensembl BioMart. Every downstream cell is unchanged.

Talking points (for explaining it)

- **Why synthetic-with-ground-truth:** it makes the notebook reproducible offline and lets you prove the scanner counts correctly before trusting it on real data.
- **Conserved core + one divergent element:** report conservation both overall ($r = 0.84$) and excluding AP-1 ($r = 0.95$); AP-1's ~3x mouse expansion is the contrast.
- **Strandedness:** real pipelines also scan the reverse complement, de-duplicating palindromic motifs such as the E-box and CRE (noted in the notebook).
- **Per-kb normalisation** so genes of different length compare fairly.
- **Dot plot vs alignment:** the dot plot shows the whole-sequence picture including indels; the pairwise alignment condenses it to a single identity number.